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Software Engineering in Industrial Practice
(SEIP)

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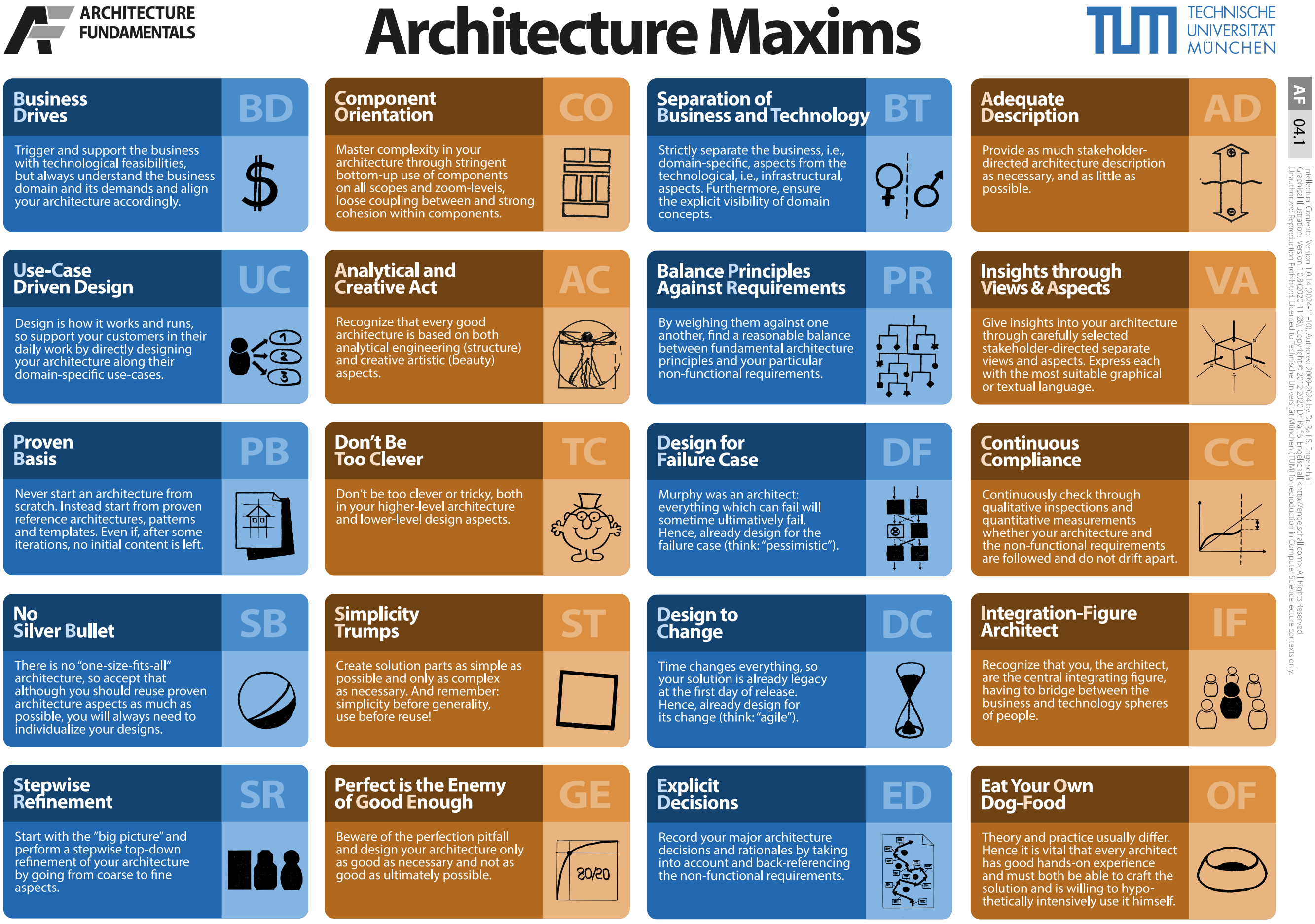

ARCHITECTURE FUNDAMENTALS

Architecture Maxims

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AF 04.1

Business Drives BD

Trigger and support the business with technological feasibilities, but always understand the business domain and its demands and align your architecture accordingly.

Component Orientation CO

Master complexity in your architecture through stringent bottom-up use of components on all scopes and zoom-levels, loose coupling between and strong cohesion within components.

Separation of Business and Technology BT

Strictly separate the business, i.e., domain-specific, aspects from the technological, i.e., infrastructural, aspects. Furthermore, ensure the explicit visibility of domain concepts.

Adequate Description AD

Provide as much stakeholder-directed architecture description as necessary, and as little as possible.

Use-Case Driven Design UC

Design is how it works and runs, so support your customers in their daily work by directly designing your architecture along their domain-specific use-cases.

Analytical and Creative Act AC

Recognize that every good architecture is based on both analytical engineering (structure) and creative artistic (beauty) aspects.

Balance Principles Against Requirements PR

By weighing them against one another, find a reasonable balance between fundamental architecture principles and your particular non-functional requirements.

Insights through Views & Aspects VA

Give insights into your architecture through carefully selected stakeholder-directed separate views and aspects. Express each with the most suitable graphical or textual language.

Proven Basis PB

Never start an architecture from scratch. Instead start from proven reference architectures, patterns and templates. Even if, after some iterations, no initial content is left.

Don't Be Too Clever TC

Don't be too clever or tricky, both in your higher-level architecture and lower-level design aspects.

Design for Failure Case DF

Murphy was an architect: everything which can fail will sometime ultimatively fail. Hence, already design for the failure case (think: "pessimistic").

Continuous Compliance CC

Continuously check through qualitative inspections and quantitative measurements whether your architecture and the non-functional requirements are followed and do not drift apart.

No Silver Bullet SB

There is no "one-size-fits-all" architecture, so accept that although you should reuse proven architecture aspects as much as possible, you will always need to individualize your designs.

Simplicity Trumps ST

Create solution parts as simple as possible and only as complex as necessary. And remember: simplicity before generality, use before reuse!

Design to Change DC

Time changes everything, so your solution is already legacy at the first day of release. Hence, already design for its change (think: "agile").

Integration-Figure Architect IF

Recognize that you, the architect, are the central integrating figure, having to bridge between the business and technology spheres of people.

Stepwise Refinement SR

Start with the "big picture" and perform a stepwise top-down refinement of your architecture by going from coarse to fine aspects.

Perfect is the Enemy of Good Enough GE

Beware of the perfection pitfall and design your architecture only as good as necessary and not as good as ultimately possible.

Explicit Decisions ED

Record your major architecture decisions and rationales by taking into account and back-referencing the non-functional requirements.

Eat Your Own Dog-Food OF

Theory and practice usually differ. Hence it is vital that every architect has good hands-on experience and must both be able to craft the solution and is willing to hypo-thetically intensively use it himself.

In der IT Architecture folgt man **Architecture Maxims**, welche grundlegende Richtlinien sind. Man kennt 20 Maxime. Der Architekt sollte den Maximen immer folgen und sie nie brechen.

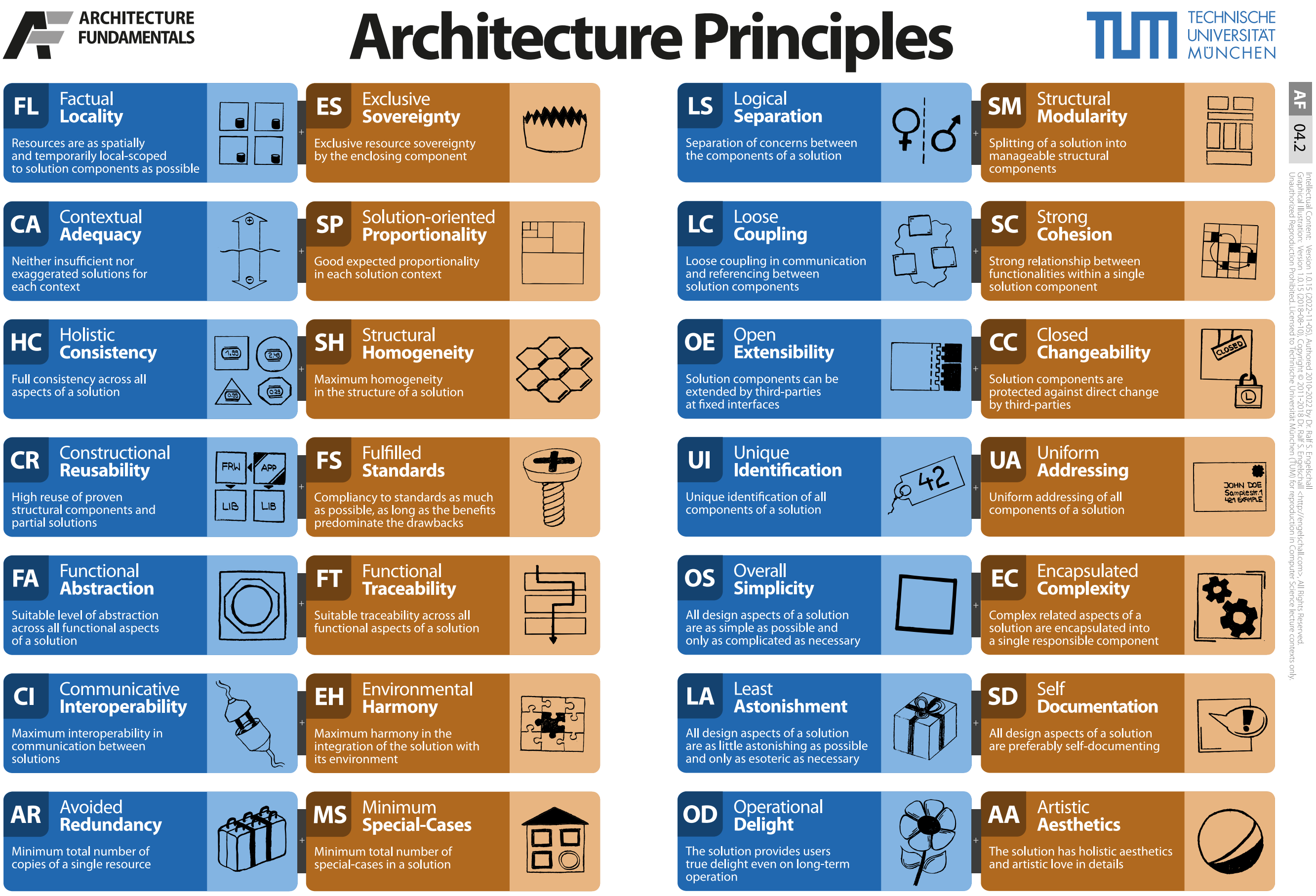
Beachte: **Proven Basis** und **No Silver Bullet** sagen, daß bei einer Architektur immer auf einer bewährten Basis (z.B. eine Referenz-Architektur) aufgesetzt werden muss, es aber gleichzeitig klar sein muss, daß man diese nicht 1:1 verwenden kann, sondern immer erst anpassen muss.

Beachte: **Stepwise Refinement** und **Component Orientation** sagen, daß vom zeitlichen Prozess her (und aus Gründen der Risikominimierung) man immer vom Groben zum Feinen geht, während die Ergebnisse eine stringente Komponenten-Orienterung aufweisen, wo kleine Komponenten hierarchisch in größeren Komponenten integriert sind.

Beachte: während man als IT-Architekt alle anderen Maxime eigentlich nur akzeptieren muss, ist **Simplicity Trumps** von einer anderen Qualität: nichts in der IT ist wirklich einfach. Wenn etwas einfach wirkt, versteht man meist nur noch nicht genügend davon. Oder jemand hat richtig viel investiert, um es einfach wirken zu lassen. **Simplicity Trumps** bedeutet genau dies zu tun: inhärent komplexe Dinge erst wieder einfach werden zu lassen.

Fragen

? Aus Gründen der Risikominimierung sollte der IT-Architekt beim **Stepwise Refinement** schrittweise immer wie vorgehen?



Architecture Principles

FL	Factual **Locality**	Resources are as spatially and temporarily local-scoped to solution components as possible
ES	Exclusive **Sovereignty**	Exclusive resource sovereignty by the enclosing component
CA	Contextual **Adequacy**	Neither insufficient nor exaggerated solutions for each context
SP	Solution-oriented **Proportionality**	Good expected proportionality in each solution context
HC	Holistic **Consistency**	Full consistency across all aspects of a solution
SH	Structural **Homogeneity**	Maximum homogeneity in the structure of a solution
CR	Constructional **Reusability**	High reuse of proven structural components and partial solutions
FS	Fulfilled **Standards**	Compliancy to standards as much as possible, as long as the benefits predominate the drawbacks
FA	Functional **Abstraction**	Suitable level of abstraction across all functional aspects of a solution
FT	Functional **Traceability**	Suitable traceability across all functional aspects of a solution
CI	Communicative **Interoperability**	Maximum interoperability in communication between solutions
EH	Environmental **Harmony**	Maximum harmony in the integration of the solution with its environment
AR	Avoided **Redundancy**	Minimum total number of copies of a single resource
MS	Minimum **Special-Cases**	Minimum total number of special-cases in a solution
LS	Logical **Separation**	Separation of concerns between the components of a solution
SM	Structural **Modularity**	Splitting of a solution into manageable structural components
LC	Loose **Coupling**	Loose coupling in communication and referencing between solution components
SC	Strong **Cohesion**	Strong relationship between functionalities within a single solution component
OE	Open **Extensibility**	Solution components can be extended by third-parties at fixed interfaces
CC	Closed **Changeability**	Solution components are protected against direct change by third-parties
UI	Unique **Identification**	Unique identification of all components of a solution
UA	Uniform **Addressing**	Uniform addressing of all components of a solution
OS	Overall **Simplicity**	All design aspects of a solution are as simple as possible and only as complicated as necessary
EC	Encapsulated **Complexity**	Complex related aspects of a solution are encapsulated into a single responsible component
LA	Least **Astonishment**	All design aspects of a solution are as little astonishing as possible and only as esoteric as necessary
SD	Self **Documentation**	All design aspects of a solution are preferably self-documenting
OD	Operational **Delight**	The solution provides users true delight even on long-term operation
AA	Artistic **Aesthetics**	The solution has holistic aesthetics and artistic love in details

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In der IT Architecture folgt man **Architecture Principles**, welche grundlegende Prinzipien und Vorgehen zusammenfassen. Man kennt 28 Prinzipien, in 14 Pärchen gruppiert werden können, da immer zwei Prinzipien inhaltlich eine starke Nähe aufweisen. Der Architekt soll den Prinzipien generell folgen, darf sie aber verletzen, solange er einen guten Grund dafür hat. Der beste Grund wäre eine spezielle projektspezifische Anforderung.

Beachte: das Prinzip **Logical Separation** (aka **Separation of Concern**) ist eines der wichtigsten, da aus ihm etliche andere Prinzipien fast automatisch folgen, unter anderem z.B. **Structural Modularity**.

Hinweis: die Prinzipien **Loose Coupling** und **Strong Cohesion** sind in der Literatur als das Kombi-Prinzip “Loose Coupling, Strong Cohesion” bekannt. Die Prinzipien **Open Extensibility** und **Closed Changeability** sind in der Literatur als das Kombi-Prinzip “Open-Close” bekannt.

Beachte: das Prinzip **Overall Simplicity** ist eines der schwersten umzusetzenden Prinzipien, da nichts in der IT wirklich einfach ist. Alles erscheint nur solange einfach, solange man noch nicht genügend davon versteht. Danach muss man es erst wieder mühsam neu “einfach” machen. Das ist die Kunst bei Architektur: schwierige Dinge zu vereinfachen! Wenn etwas nicht viel weiter vereinfacht werden kann und immer noch gewisse Komplexität aufweist, kann man es zumindest mit dem Prinzip **Encapsulated Complexity** versuchen zu verschatten.

Fragen

- ? Zählen Sie mindestens 4 wesentliche **Architecture Principles** auf!

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Component Design

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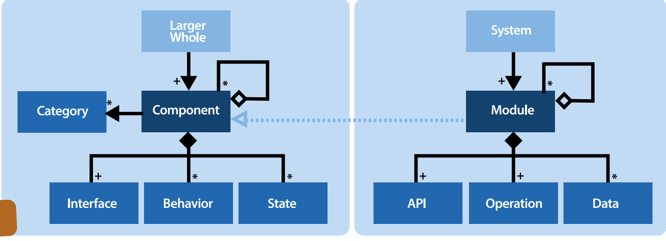
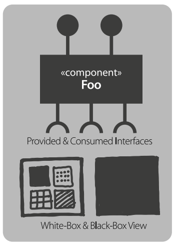
Definition of a Component (of a Larger Whole):
a **know-how encapsulating**, potentially **reusable** and **substitutable** **unit of hierarchical composition** with **explicit context dependencies**, which **hides the complexity** of its optional **behavior** and **state** realization behind **small contractually specified interfaces**, defines its **added value** in terms of **provided** and **consumed** interfaces and optionally belongs to zero or more **categories** of similar units.

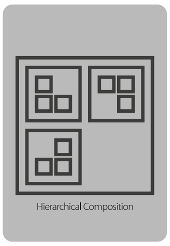
Definition of a Module (of a System):
a know-how encapsulating, potentially reusable and substitutable **source-code** unit of hierarchical composition with explicit context dependencies which hides the complexity of its **operation** and **data implementation** behind small contractually specified **Application Programming Interfaces (API)**, defines its added value in terms of provided and consumed **APIs** and optionally belongs to zero or more categories of similar units.

Example Categories of Components:
- Namespace
- Directory, File
- Configuration, Section, Directive
- Host, Virtual Machine, Container
- Process Group, Process, Thread
- Application, Microservice, Program
- Package, Class, Function
- Database, Schema, Table, Record
- Datamodel, Entity Group, Entity
- User Interface, Dialog, Widget

Any group of anything!







How to find Components (or Modules)?

DCA Domain Concept Abstraction
Model domain concepts as entity components and then group at higher levels.

SOC Separation of Concerns
Build components for clearly distinct concerns.

USE Reusability Potential
Decide on components based on their reusability potential.

UCC Use-Case Clustering
Build domain components for each use-case or each logical use-case cluster.

SRP Single Responsibility Principle
Build components for clearly distinct responsibilities.

DCC Divide & Conquer Complexity
Master overall complexity by splitting larger things into smaller things.

DDD Domain-Driven Design
Model domain "Bounded Contexts" through DDD and derive components from them.

CNC Coupling and Cohesion
Decide on components based on their loose coupling and strong cohesion.

CCC Cross-Cutting Concerns
Build common cross-cutting concerns as cross-cutting components.

OOD Object-Oriented Design
Model Object-Oriented Design entities (and/or OOP constructs) as components.

DEP Dependency Encapsulation
Decide on components based on their encapsulation of dependencies.

PAT Architecture Patterns
Build inner components to comply to outer structure, slicing and clustering architectures.

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In der Software Architecture dreht sich alles um Komponenten (**Components**) und Schnittstellen (**Interfaces**). Deshalb ist **Component Design** eine zentrale Aufgabe des Architekten.

Eine Component **kapselt** ein bestimmtes **Know-How**, ist **potenziell wiederverwendbar** und **ersetzbar**. Eine Component besitzt ein **Verhalten** und einen **Zustand** und verbirgt die interne Komplexität von beidem hinter "kleinen" **vertraglichen Schnittstellen**. Sie stellt ihren Mehrwert über die Differenz bereitgestellter und konsumierter Schnittstellen bereit. Sie kann als **Whitebox** oder **Blackbox** betrachtet werden, abhängig davon, ob man ihre internen Details von außen sieht oder nicht. Components sind hierarchisch angeordnet, gehören eventuell zu bestimmten **Kategorien** und besitzen **explizite Abhängigkeiten** untereinander.

Man unterscheidet zwischen dem allgemeineren Konzept der **Component** ("any group of anything") und dem spezielleren Konzept des (über Source Code definierten) **Module**.

Components findet man über zahlreiche Wege. Die meisten leiten sich direkt aus bestehenden Methoden, Prinzipien oder Mustern ab. Die beiden wichtigsten Wege für einen Komponentenschnitt in der Praxis sind: **Separation of Concerns** (welches einmalige Anliegen bzw. welche einmalige Aufgabe hat die Komponente?) und **Single Responsibility Principle** (welche einmalige Verantwortung hat die Komponente?).

Fragen

- Zählen Sie mindestens 7 Eigenschaften/Aspekte auf, die eine Komponente besitzt!
- Was sind die beiden wichtigsten Wege, um in der Praxis einen Komponenteschnitt zu finden?

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Interface Design

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AF 05.2

Definition of an Interface: well-defined shielding and abstracting **boundary** of a passive, providing **component**, consisting of one or more distinguished, **outside-in** designed, **interaction endpoints**, each accessed and controlled by active, consuming components through the **exchange** of **input/output information** and operating under a certain **syntactical** and **semantical contract.**

Endpoint: Name, Directive, Command, Function, Method, Procedure, Address, Port, URL, Dialog, ...
Exchange: Option, Argument, Parameter, Return Value, Result, Request/Response Message, Error/Exception, Interaction, ...
Contract: Syntax, Pre-Condition, Invariant, Post-Condition, Side-Effect, Idempotence, Determinism, Functionality, ...

Types of Software Interfaces

API **Application Programming Interface**
Example: `foo("bar", 42)` (call and use)

SPI **Service Provider Interface**
Example: `register("foo", (x,k) => ...)` (extend and implement)

SCI **Startup Configuration Interface**
Examples: INI, Java Properties, TOML, YAML, JSON, XML, etc.

BPI **Batch Processing Interface**
Examples: Unix at(1), Unix ts(1), GNU Batch, Spring Batch, Java Batch, SAP LO-BM, etc.

CLI **Command-Line Interface**
Example: `foo -x --bar=baz quux`

GUI **Graphical User Interface**
Examples: Windows/WPF, macOS/Cocoa, KDE/Qt, GNOME/GTK

RNI **Remote Network Interface**
Examples: GraphQL-IO, HTTP/REST, SOAP, RMI, WebSockets, AMQP, MQTT, etc.

Characteristics of Good Interfaces

AP **Appropriate** & **Proportional**
Appropriate to consumer requirements, proportional to provider functionality.

SA **Shielding** & **Abstracting**
Shields from direct access, abstracts and hides implementation details.

IE **Inviting** & **Expressive**
Invites through "outside – in" design, powerful in expressiveness.

IF **Intuitive** & **Foolproof**
Intuitive to grasp and use, hard to misuse.

OC **Orthogonal** & **Concise**
Supports combinatorial use-cases, causes minimum boilerplate.

TP **Tolerant** & **Predictable**
Tolerant on input, predictable on output.

EC **Extensible** & **Compatible**
Easy to extend for providers, backward/forward-compatible for consumers.

Selected Interface Design Patterns

IVF **Interface Version & Features**
Provide version and feature information for algebraic comparison and feature detection.

2LF **Leaky Two-Layer Facade**
Provide higher-level convenient use-case and lower-level orthogonal feature interface.

EVE **Event Emitter**
Emit events to previously registered, interested consumers.

CTX **Multi-Context**
Use contexts to distinguish between different usage scenarios and to carry common info.

CEF **Configure-Execute Flow**
Spread use-cases onto a flow of configuration exchanges and a final executional exchange.

IOC **Inversion Of Control**
Invert control on asynchronous operations via callbacks, promises or async. mechanisms.

HMR **Human/Machine Responses**
Support humans and machines in outputs through both description and parsing-free info.

Eine Schnittstelle (**interface**) ist eine wohldefinierte (**well-defined**), abschirmende (**shielding**), abstrahierende (**abstracting**) Berandung (**boundary**) einer passiven bereitstellenden Komponente (**component**), welche aus einer oder mehreren klarunterscheidbaren Interaktionsendpunkten (**interaction endpoints**) besteht.

Auf jeden Interaktionsendpunkt wird dabei von einer aktiven konsumierenden Komponente durch den Austausch (**exchange**) von **input/output** Informationen (**information**) zugegriffen und wird unter einem bestimmten syntaktischen (**syntactical**) und semantischen (**semantical**) Vertrag (**contract**) betrieben.

Es gibt zahlreiche Arten von Schnittstellen, die alle dieser Definition entsprechen. "Gute" Schnittstellen haben darüber hinaus bestimmte Eigenschaften/Charakteristiken. Die vier der besten Eigenschaften sind: **Proportional** (die Schnittstelle ist kleiner und in der Größe proportional zur dahinterliegenden Funktionalität), **Expressive** (die Schnittstelle stellt ein leistungsfähiges Programmiermodell zur Verfügung), **Orthogonal** (die Schnittstelle erlaubt kombinatorische Use-Cases), und **Concise** (die Schnittstelle erzeugt wenig "Boilerplate Code" bei der Nutzung).

Es gibt zahlreiche Software Pattern für Schnittstellen. Ein interessantes Muster ist die **Leaky Two-Layer Facade**, bei dem eine Bibliothek zwei Schnittstellen besitzt: eine obere, bequeme und Use-Case-bezogene Schnittstelle und eine untere, orthogonale Feature-bezogene Schnittstelle. Der Witz ist, daß die obere durch alleine die untere Schnittstelle implementiert wird und daß die untere Schnittstelle "durchscheint" ("leaky" bzw. Open Layering).

Fragen

- ❓ Zählen Sie mindestens 8 Eigenschaften/Aspekte auf, die eine Schnittstelle definieren!
- ❓ Zählen Sie mindestens 4 Eigenschaften/Charakteristiken von guten Schnittstellen auf!